

***TEDI* user guide: an interactive inventory of toxicological and exposure data sources**

1. Problem

How can toxicological and exposure data be found?

2. Solution: method and approach

Knowledge of hazards and exposure data for chemical pollutants is essential for epidemiological studies aimed at assessing health risks and preventing disease development. Access to this type of information is clearly necessary, yet although numerous data sources have been compiled, they are often extremely difficult to identify.

A literature review was conducted to collect metadata on toxicological and exposure data sources that can be used for chemical exposome health risk assessment. The results of this bibliographic search have been compiled into a free online inventory called *TEDI* (*Toxicological and Exposure Database Inventory*) [Petit, 2022].

TEDI is available free of charge at <https://exporisk-timc.imag.fr/TEDI/> and is regularly updated. In total, *TEDI* identifies, catalogs, and describes 1,085 data sources. The number of chemical substances ranges from a single chemical to 90 million for toxicological data sources and from 1 to 103,817 chemicals for exposure data sources. Some data sources are limited to specific chemicals, with for example, 112 exposure data sources report polycyclic aromatic hydrocarbon (PAH) measurements. *TEDI* is organized into 8 tabs (Figure 1). Local searches (i.e., by column) or global searches can be performed. All tables are interactive and fully downloadable. Orange underlined terms are hyperlinks that can be clicked to access definitions or be directed to the data source.

The first tab lists toxicological data sources (n=456), which provide a wide range of information such as toxicity data (57%), physicochemical properties (38%), experimental study results (28%), and predictive data (16%). Toxicological reference values (TRVs) are also available. The second tab lists exposure data sources (n=668), including exposure measurement data (71%), web applications/tools (21%), exposure and risk assessment models (19%), and epidemiological data sources (13%). The third tab lists exposure data sources reporting PAH measurements (n=112). The fourth tab provides the data underpinning *TEDI*. Finally, the last four tabs provide information on *TEDI*'s use in the scientific literature, available help, information about *TEDI*, and legal notices.

 TEDI - Toxicological and Exposure Database Inventory						
Toxicological databases Exposure databases PAH databases References News & Articles Help/Info Contact Mentions légales						
There are currently a total of 112 exposure measurement databases regarding polycyclic aromatic hydrocarbons (PAHs) listed in TEDI.						
Tab. 3: Inventory of exposure databases regarding PAHs Copy CSV Excel PDF Print Column visibility Show 10 rows Search:						
Name of the database	Pollutant containing or emitting PAHs	Number of PAHs	PAH recorded	Number of biomarkers	Biomarker recorded	Reference
All	All	All	All	All	All	All
Air.Q+	Not concerned	1	BaP	0	Not concerned	Hassan Bhat et al. 2021
AirBase	Not concerned	7	BaA , BaP , BbF , BjF , BkF , DahA , IP	0	Not concerned	Guerreiro et al. 2016 ; Mol and van Hooydonk. 2013
AQ e-Reportline	Not concerned	1	BaP	0	Not concerned	EEA. 2016
AQS	Not concerned	16	Naoh , Acen , Acen , Fluo , Pyr , BghiP , BaA , Chr , BaP , IP , Phe , Flua , Ant , BbF , BkF , DahA	0	Not concerned	Hirabayashi et al. 2016 ; Liu et al. 2017
ASA register	Not concerned		PAH	0	Not concerned	Brooke et al. 2006 ; Kauopinen et al. 2007 ; Lissner et al. 2014
ATABAS	Not concerned	1	Naoh	0	Not concerned	Brooke et al. 2006 ; Hansen et al. 1993 ; Jensen et al. 1990 ; Simonsen and Lund. 1992
ATHLETE	Not concerned	0	Not concerned	18	1-OHP , 3-OHBP , other PAH metabolites not reported	Vrijheid et al. 2021
Atmo Auvergne Rhone-Alpes	Not concerned	1	BaP	0	Not concerned	Aix et al. 2022
AWE	Bitumen fume, bitumen vapour, diesel exhaust		PAH	0	Not concerned	Burstyn et al. 2000a
AWES	Coal-tar pitch, creosotes, frying emission from high temperatures, mineral oils (treated or mildly treated), soots, diesel engine exhaust	3	BaP , CcdP , DahP	0	Not concerned	Carev et al. 2014 ; Driscoll et al. 2016 ; Lissner et al. 2014
Note: empty cells correspond to non-reported values. Orange font refers to hyperlinks that provide both definition and description of terms or direct access to the mentioned references.						
Showing 1 to 10 of 112 entries				Previous 1 2 3 4 5 ... 12 Next		

Figure 1: TEDI overview

3. Key points

TEDI goes beyond a simple inventory by providing a user-friendly tool that guides risk assessors, risk managers, practitioners, decision-makers, and health agencies in identifying, locating, and accessing information essential for conducting health risk assessments and epidemiological studies.

4. Reference

Petit P. Toxicological and Exposure Database Inventory: a review. *Int J Hyg Environ Health*. 2022;246:114055. <https://doi.org/10.1016/j.ijheh.2022.114055>.